Math 534H Lec 15

Stability ~~Stability~~ of solutions to Heat/Diffusion equation

Consider two solution to the IVBP for H/D equation

$$\begin{cases} \partial_t u_1 - k\partial_{xx} u_1 = 0, & \boxed{x \in [0, L] \quad t > 0} \\ u_1(x, 0) = \phi_1(x) \\ u_1(0, t) = g_1(t), & \boxed{u_1(L, t)} = h_1(t) \end{cases}$$

$$\begin{cases} \partial_t u_2 - k\partial_{xx} u_2 = 0 \\ u_2(x, 0) = \phi_2(x) \\ u_2(0, t) = g_2(t) \qquad u_2(L, t) = h_2(t) \end{cases}$$

Assume: $\phi_1 \xrightarrow{\text{closeness}} \phi_2 \qquad g_1 \xrightarrow{\text{closeness}} g_2$

$$u_1 \xrightarrow{\text{close}} u_2$$

Consider $\quad W = u_1 - u_2$

$$\begin{cases} \partial_t W - k\partial_{xx} W = 0 \\ W(x, 0) = \cancel{\phi W(x)}\ \phi_1(x) - \phi_2(x) \\ W(0, t) = g_1(t) - g_2(t) \\ \qquad\qquad W(L, t) = h_1(t) - h_2(t) \end{cases}$$

$\downarrow$ By max/min principle imply:

$$\min_{(x,t)\in I} \{W(x,t)\} \le \underbrace{W(x,t)}_{(x,t)\in R} \le \max_{(x,t)\in I} \{W(x,t)\}$$

Infact: ① $-\max_{(x,t)\in I} \{|W(x,t)|\} \le \max_{(x,t)\in R} \{W(x,t)\} \le \max_{(x,t)\in I} \{|W(x,t)|\}$

$\Downarrow$

② $\max_{(x,t)\in R} \{|u_1(x,t) - u_2(x,t)|\} \le \max_{(x,t)\in I} \{|W(x,t)|\}$

$= \max_{\cancel{0\le x\le L}\ (x,t)\in I} \{|\phi_1(x) - \phi_2(x)|, |g_1(t) - g_2(t)|, |h_1(t) - h_2(t)|\}$

$\boxed{\text{Uniformly}}$

Hence if the maximum of $|\phi_1(x) - \phi_2(x)|$ and the maximum of $|g_1(t) - g_2(t)|$ and of $|h_1(t) - h_2(t)|$ are small then maximum of $|u_1(x,t) - u_2(x,t)|$ is also small.

measure stability be square ability by the energy functional.

Claim $E[u](t)$ is decreasing.

$$\underline{0 \le E[u](t) \le E[u](0)} \qquad \forall \underline{u}_{(\text{solution})}$$

$$t > 0$$

The energy of the solution is controlled by $\underbrace{E[|\phi_1 - \phi_2|]}_{= E[w](0)}$

In fact, we will proof the energy for the 4/D equation is nothing but:

$$E[u](t) = \frac{1}{2}\int_0^L |u(x,t)|^2 dx$$

and $\frac{dE}{dt} \le 0$ so energy decrease.

$$\int_0^L |u_1(x,t) - u_2(x,t)|^2 dx \le \underbrace{\int_0^L |\phi_1(x) - \phi_2(x)|^2 dx}_{\text{number.}}$$

This type of stability is called in the square integrab sense.

Resistence Term / Forcing term

Energy functional for H/D equations.

$$0 = u_{tt} - k u_{xx}$$
$$u \cdot 0 = u_t u - k u_{xx} u$$
$$\boxed{0 = u u_t - k u u_{xx}}$$

下文的 w 替换 u

$$\begin{cases} w_t - k w_{xx} = 0 \\ w(x,0) = 0 \\ w(0,t) = 0 \\ w(L,t) = 0 \end{cases} \quad x \in [0, L]$$

$\cancel{w_x}$

$$\begin{cases} 0 = w_t - k w_{xx} \\ w \cdot 0 = (w_t - k w_{xx}) w = 0 \\ 0 = w w_t - k w w_{xx} \end{cases}$$

$$0 = \underline{\int_0^L w w_t \, dx} - \underline{k \int_0^L w w_{xx} \, dx}$$

$$\underline{\int_0^L \frac{1}{2} \frac{d}{dt} (w^2) \, dx}$$

$$\underbrace{-k \left[w w_x \right]_{x=0}^{x=L}}_{=0} - \boxed{k \int_0^L (w_x)^2 \, dx}$$

Hence:

$$\frac{1}{2} \frac{d}{dt} \int_0^L |w(x,t)|^2 = \underbrace{-k \int_0^L (w_x)^2 \, dx}_{\leq 0}$$

$$\underline{E[w](t) = \frac{1}{2} \int_0^L |w(x,t)|^2}$$

$$\frac{dE}{dt} \leq 0 \quad \boxed{\text{decrease.}}$$

Uniqueness: u_1 and u_2 $\longleftarrow$ solution (IVBP)

$$\Downarrow$$

$\underline{w = \cancel{u} \; u_1 - u_2}$ solve. IVBP

$$\Big\downarrow \text{Energy decrease.}$$

$$E[w](t) \le E[w](0)$$

$$\Big| \le \frac{1}{2}\int \underbrace{|w(x,0)|^2 dx}_{=0}$$

$$\Big\downarrow$$

$$0 \le E[w](t) \le 0$$

$$\Downarrow$$

$$0 \le \underbrace{\int_0^L |w(x,t)|^2 dx}_{} \le 0$$

$$\Downarrow$$

$$0$$

$$\cancel{w(x)} \Downarrow$$

$$\underline{w(x,t) = 0} \quad \forall x. \ \forall t$$

$$\Downarrow$$

$$\boxed{w = 0 = \overset{u_1 - u_2}{\cancel{u_1 - u_2}}} \Rightarrow \boxed{u_1 = u_2}$$